

ZAINEB DARCHEM

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OBJECTIVE

Aspiring AI Engineer with a solid foundation in applied mathematics, statistics and data science, eager to apply advanced ML techniques to solve business challenges. Skilled in Python programming, model development and deployment, with hands-on project experience in agentic systems, predictive modeling and recommenders.

EXPERIENCE

Banque Nationale Agricole (BNA) : Accounting Reporting Division Intern August 2025

- Automated Excel-based accounting workflows using Python, VBA, and Power BI M language.
- Built scripts to combine, clean, and transform financial datasets, reducing processing time from minutes to seconds.
- Integrated automated data preparation directly into Power BI for seamless dashboard refreshes.

EDUCATION

Applied Mathematics and Modeling Engineering 2024 - Present
Higher National Engineering School of Tunis (ENSIT)

Maths & Physics Preparatory degree : 2022 - 2024
Preparatory Institute for Engineering Studies of Tunis (IPEIT)

SKILLS

- Programming:** Python, SQL, java, C, C++, MATLAB, HTML/CSS, Javascript
- ML & AI:** Langgraph, Langchain, Scikit-Learn, Pandas, TensorFlow, PyTorch, NumPy, Matplotlib, Seaborn
- Mathematics:** Linear algebra, probability distributions, statistical inference, optimization
- Tools/Environments:** Git/GitHub, VSCode, jupyter, google colab

PROJECTS

AutoInsight: LLM-Powered Multi-Agent System for Automated Machine Learning Pipelines

- Built a LangGraph-based multi-agent AutoML system with LLM-powered agents, orchestrated via a central controller to manage end-to-end workflow (EDA, preprocessing, feature engineering, modeling)
- Developed an NLP parser using LLMs to extract user intent (task type) and automatically identify the target column, enabling fully automated pipeline configuration from natural language input.
- Implemented end-to-end automation with validation and reporting, including pipeline consistency checks and generation of structured final reports summarizing data insights, modeling decisions, and performance metrics.

Healthcast : AI-driven Fitness & Nutrition App ([Try it here](#))

- Built an app that converts natural-language health and lifestyle prompts into personalized workout and meal plans.
- Collaborated in a team, sharing roles across NLP extraction, ML models training and generative AI integration.
- Implemented Random Forest and XGBoost models for workout and nutrition predictions using custom datasets.
- Integrated LLMs to produce motivational podcast-style audio and concise Markdown summaries for users.
- Developed a Streamlit web interface for intuitive user input and output interaction.

COMMUNITY INVOLVEMENTS

- 8th Global Competition on Design for Future Education (GCD4FE) in China**
- Seneca Hacks : 48h AI hackathon**
- NerData ENSIT Club :** Project manager organizing hackathons & conducting various AI workshops
- Geeks ENSIT Club :** Active Member

ADDITIONAL QUALIFICATIONS

Languages : english (advanced), french (fluent), arabic (native), german (beginner)

Certificates :

- Designing Agentic Systems with LangChain - datacamp
- Mathematics for ML - 365 DataScience
- Hands on TensorFlow - Pistacia